Module 4- Task 3

Task A (Set): Update a specific product to change its price.

1. Updating price of “Ball Pen Pack” to 210 Query :

```
db.products.updateOne(
 {name: "Ball Pen Pack"},
 { $set: {price: 210}}
)
```

Explanation:

This query updates the price of the product "Ball Pen Pack" to 210 using $set. It modifies only the specified field without affecting other fields in the document.

```
> db.products.findOne(
    {name: 'Ball Pen Pack'},
    {"_id":0, "name": 1, "price":1
  })
<
    name: 'Ball Pen Pack',
    price: 199
  }
> db.products.updateOne(
    {name: "Ball Pen Pack"},
    { $set: {price: 210}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.products.findOne(
    {name: 'Ball Pen Pack'},
    {"_id":0, "name": 1, "price":1
  })
<
    name: 'Ball Pen Pack',
    price: 210
  }
```

2. Updating price of all products for category = stationery with value= original value + 100

   Query:

```
db.products.updateMany(
 { category: "Stationery" },
 { $inc: { price: 100 } }
)
```

Explanation:
This query increases the price of all products in the "Stationery" category by 100 using $inc. It allows batch updates for multiple documents that meet the filter condition.

```
> db.products.find(
    {category: "Stationery"},
    {"_id": 0, "name": 1, "price": 1}
  )
< {
    name: 'Notebook Pack',
    price: 300
  }
  {
    name: 'Notebook Set',
    price: 300
  }
  {
    name: 'Ball Pen Pack',
    price: 300
  }
```

```
> db.products.updateMany(
    { category: "Stationery" },
    { $inc: { price: 100 } }
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
> db.products.find(
    {category: "Stationery"},
    {"_id": 0, "name": 1, "price": 1}
  )
< {
    name: 'Notebook Pack',
    price: 400
  }
  {
    name: 'Notebook Set',
    price: 400
  }
  {
    name: 'Ball Pen Pack',
    price: 400
  }
```

3. Renaming the name field of Goggles to Product name Query:

```
db.products.updateOne(
 {_id: ObjectId('69723a04c3edee440f521909')},
 {$rename: {"name": "Product name"}}
```

Explanation:
This query renames the name field of a specific product (identified by _id) to "Product name" using $rename. It changes the field key while keeping the existing data intact.

```
> db.products.find(
    {name: "Goggles"}
  )
< {
    _id: ObjectId('69723a04c3edee440f521909'),
    name: 'Goggles',
    category: 'Accesories',
    price: 500,
    tags: [
      'Beauty',
      'Daily fashion'
    ]
  }
```

```
> db.products.updateOne(
    {_id: ObjectId('69723a04c3edee440f521909')},
    {$rename: {"name": "Product name"}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.products.find(
    {'Product name': "Goggles"}
  )
< {
    _id: ObjectId('69723a04c3edee440f521909'),
    category: 'Accesories',
    price: 500,
    tags: [
      'Beauty',
      'Daily fashion'
    ],
    'Product name': 'Goggles'
  }
```

Task B (Push Array): Add a new tag "Best-seller" to the existing Tags array for a specific Product.

1. Adding Best Seller tag to a product named: Ball Pen Pack

Query:

```
db.products.updateOne(
  {name: "Ball Pen Pack"},
  { $push: {tags: "Best Seller"}}
)
```

Explanation:
This query adds "Best Seller" to the tags array of the product "Ball Pen Pack" using $push. It appends the value to the end of the array.

```
> db.products.findOne
  (
    {name: "Ball Pen Pack"}
  )
< {
    _id: ObjectId('69721e2313c4d8697d25d28b'),
    name: 'Ball Pen Pack',
    category: 'Stationery',
    price: 400,
    stock: 300,
    tags: [
      'pen',
      'writing',
      'office'
    ]
  }
```

```
>  db.products.updateOne(
     {name: "Ball Pen Pack"},
     { $push: {tags: "Best Seller"}}
   )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.products.findOne
  (
    {name: "Ball Pen Pack"}
  )
< {
    _id: ObjectId('69721e2313c4d8697d25d28b'),
    name: 'Ball Pen Pack',
    category: 'Stationery',
    price: 400,
    stock: 300,
    tags: [
      'pen',
      'writing',
      'office',
      'Best Seller'
    ]
  }
salessight_db >
```

2. Now remove that Best Seller tag using $pull

Query:

```
db.products.updateOne(   {name: "Ball Pen Pack"},
  { $pull: {tags: "Best Seller"}}
)
```

Explanation:
This query removes the "Best Seller" tag from the tags array using $pull. It deletes any array element matching the specified value.

```
> db.products.updateOne(
    {name: "Ball Pen Pack"},
    { $pull: {tags: "Best Seller"}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.products.findOne
  (
    {name: "Ball Pen Pack"}
  )
< {
    _id: ObjectId('69721e2313c4d8697d25d28b'),
    name: 'Ball Pen Pack',
    category: 'Stationery',
    price: 400,
    stock: 300,
    tags: [
      'pen',
      'writing',
      'office'
    ]
  }
```

3. Use $pop to exatract the last tag (1 for last -1 for first)

Query:

```
db.products.updateOne(
  {name: "Ball Pen Pack"},
  { $pop: {tags: 1}}
)
```

Explanation:
This query removes the last element of the tags array for "Ball Pen Pack" using $pop: 1. $pop allows removing either the first (-1) or last (1) element in an array.

```
> db.products.updateOne(
    {name: "Ball Pen Pack"},
    { $pop: {tags: 1}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.products.findOne
  (
    {name: "Ball Pen Pack"}
  )
< {
    _id: ObjectId('69721e2313c4d8697d25d28b'),
    name: 'Ball Pen Pack',
    category: 'Stationery',
    price: 400,
    stock: 300,
    tags: [
      'pen',
      'writing'
    ]
  }
```

4. Now Use $addToSet to add that deleted tags again:

Query:

```
db.products.updateOne(
 {name: "Ball Pen Pack"},
 {$addToSet: {tags: {$each: ["obice", "Best Seller"]}}}
)
```

Explanation:
This query adds "obice" and "Best Seller" back to the tags array using $addToSet with $each. It ensures no duplicate entries are added to the array.

```
> db.products.updateOne(
    {name: "Ball Pen Pack"},
    {$addToSet: {tags: {$each: ["office", "Best Seller"]}}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.products.findOne
  (
    {name: "Ball Pen Pack"}
  )
< {
    _id: ObjectId('69721e2313c4d8697d25d28b'),
    name: 'Ball Pen Pack',
    category: 'Stationery',
    price: 400,
    stock: 300,
    tags: [
      'pen',
      'writing',
      'office',
      'Best Seller'
    ]
  }
```